

BASIC LEVEL QUESTIONS & ANSWERS

1. What is the average rating of apps in the dataset?

Answer: Average app rating is approximately **4.2**.

2. How many unique categories of apps are there?

Answer: There are **33 unique app categories**.

3. What is the distribution of app sizes?

Answer: Most apps are under **20 MB**, with very few apps exceeding **100 MB**. The distribution is right-skewed.

4. How many free vs paid apps are there?

Answer: Free apps dominate the Play Store, representing over **90%** of total apps, while paid apps form a small portion.

5. What is the most common content rating?

Answer: **Everyone** is the most common content rating category.

6. What are the top 5 most installed apps?

Answer:

1. Facebook
2. Gmail
3. Google
4. Google Chrome
5. Google Drive

(All have around **1 Billion installs**.)

7. How many apps have a rating of 4.0 and above?

Answer: Approximately **6,285 apps** have ratings of 4.0 or higher.

8. What is the average number of reviews for free vs paid apps?

Answer: Free apps receive significantly more reviews on average than paid apps because of their larger user base.

9. What is the average app size for each category?

Answer: Categories such as Games and Video Players tend to have larger average sizes, while Education and Productivity apps are generally smaller.

10. How many apps were last updated in 2018?

Answer: Most apps in the dataset were updated during 2018, indicating active maintenance by developers.

MEDIUM LEVEL QUESTIONS & ANSWERS

1. What is the correlation between installs and ratings?

Answer: There is a **weak positive relationship** between installs and ratings. Popular apps generally maintain good ratings, but installs alone do not guarantee higher ratings.

2. Which categories have the highest average ratings?

Answer:

- Events
- Education
- Art & Design
- Books & Reference
- Personalization

These categories show average ratings above **4.3**.

3. How does price affect average rating?

Answer: Paid apps generally have slightly higher ratings than free apps, suggesting users expect and receive better quality from paid applications.

4. What is the distribution of ratings across content ratings?

Answer: All content-rating groups maintain average ratings between **4.1 and 4.3**, showing little variation.

5. Which genres have the most apps with over 1 million installs?

Answer:

- Communication
- Game
- Social
- Tools
- Productivity

These genres dominate the high-install segment.

6. How frequently do apps get updated?

Answer: Most successful apps are updated regularly. Frequent updates help maintain ratings and user engagement.

7. What is the impact of app size on installs?

Answer: App size has only a minor effect on installs. Large apps can still achieve very high download counts if they deliver strong value.

8. Which apps have the highest number of reviews and what are their ratings?

Answer:

App	Reviews	Rating
Facebook	78M+	~4.1
WhatsApp Messenger	69M+	~4.4
Instagram	66M+	~4.5
Messenger	56M+	~4.0
Clash of Clans	45M+	~4.6

9. How does content rating distribution differ between free and paid apps?

Answer: Both free and paid apps are mainly targeted toward the **Everyone** audience, although paid apps show slightly more diversity in content ratings.

10. What are the top 5 categories with the most installs?

Answer:

1. Game
2. Communication
3. Tools
4. Productivity
5. Social

These categories account for the largest share of downloads.

ADVANCED LEVEL QUESTIONS & ANSWERS

1. What are the top 10 apps with the highest ratings, and how do their reviews and installs compare?

Answer: Several apps achieve perfect or near-perfect ratings, but many have relatively fewer reviews and installs compared with global giants like Facebook and Instagram. High ratings do not always mean high popularity.

2. Analyze the trend of app updates over time.

Answer: Update frequency increased significantly in recent years, with 2018 showing the highest concentration of updates. This suggests developers actively maintain their apps to remain competitive.

3. How does average rating change with installs? (Binned Analysis)

Answer: Ratings remain relatively stable across install ranges, generally between 4.0 and 4.5. Extremely high install counts do not necessarily lead to higher ratings.

4. Perform sentiment analysis on app reviews.

Answer: Review text is not available in the dataset, so sentiment analysis could not be performed. If review text were available, themes such as performance, usability, and customer support could be analyzed.

5. What is the relationship between genre and user ratings?

Answer: Genres such as:

- Events
- Education
- Art & Design
- Books & Reference

tend to achieve higher ratings, while some entertainment-focused genres show slightly lower average ratings. User satisfaction varies by genre but generally remains above 4.0.